

Zihui Zhou

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Github: <https://zhouzihui2001.github.io/>

Research Interests

Multi-modal Representation Learning, Vision-Language Models, Computer Vision, Remote Sensing AI

Education

Chongqing University (Project 985, 211)

MSc in Computer Science and Technology

Sep. 2023-Present

- Major: Computer Science and Technology
- Weighted average score: **92.1/100.0**

Chongqing University (Project 985, 211)

BSc in Computer Science and Technology

Sep. 2019-Jun. 2023

- Major: Computer Science and Technology
- Weighted average score: **88.6/100.0**

Research Experience

Assessment and Mitigation of Hallucinations in Multimodal Large Language Models within the Domain of Remote Sensing

(Oct. 2024-Now)

- Characterized multimodal large language model (MLLM) hallucination patterns in the remote sensing field, establishing a taxonomy that categorizes 4 different hallucination types.
- Proposed an MLLM hallucinations detection suite, covering the RSHalluEval benchmark, evaluation metrics, and flexible automated detection strategies for different scenarios, to assess MLLM hallucinations in this domain.
- Presented RSHalluShield - a 30,000 QA-pair dataset for hallucination mitigation.
- Proposed two effective plug-and-play hallucination relief schemes for training-free situations.
- Main contributions: project overall design, implementation, and evaluation of the entire project
- Submitted to *IEEE Transactions on Geoscience and Remote Sensing* (Q1, IF:8.2), under review (first author).

Fine-Grained Information Supplementation and Value-Guided Learning for Remote Sensing Image-Text Retrieval

(Nov. 2023-Aug. 2024)

- Proposed the fine-grained information supplementation (FGIS) module: enhancing the model's perception of multi-scale features in remote sensing images by fusing global and local visual features.
- Crafted the value-guided learning framework (VGLF), including a weighted contrastive loss, scene-adaptive fine-grained perceptual loss, and two additional losses. Implemented targeted training-phase strategies to prioritize high-value data during training.
- Main contributions: project overall design, implementation, and evaluation of the entire project
- One paper accepted by the *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* (first author, <https://ieeexplore.ieee.org/document/10716520>).

Project: Intelligent Environmental Monitoring System Based on Multi-source Satellite Remote Sensing Images

(Sep. 2023-Jun. 2024)

- Developed a functional module for high-resolution remote sensing image segmentation, capable of segmenting objects of 13 categories in images of different sizes.
- Deployed the remote sensing MLLM GeoChat to implement intelligent remote sensing question-answering.

Bachelor Thesis: Research and Implementation of Image Text Retrieval Based on Attention Mechanism

(Oct. 2022-May 2023)

- Developed an intra-modal loss function with three forms (visual, text, and combined modalities) to constrain global feature disparities between positive instances and hard negatives, thus obtaining superior global feature representation.
- Conducted experiments to demonstrate the superiority of the proposed model and developed a prototype image-text retrieval system based on it.

Publications

[1] **Zhou, Z.**, Feng, Y., Qiu, A., Duan, G., & Zhou, M. (2024). Fine-Grained Information Supplementation and Value-Guided Learning for Remote Sensing Image-Text Retrieval. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.

[2] **Zhou, Z.**, Feng, Y., Duan, G., Zhou, M., & Jia, W. (Under Review). Assessment and Mitigation of Hallucinations in Multimodal Large Language Models within the Domain of Remote Sensing. *IEEE Transactions on Geoscience and Remote Sensing*.

Internship Experience

Chongqing Longline Intelligent Technology Co., Ltd., Perception and Localization Department

Algorithm Engineer

Dec. 2024-Apr. 2025

- Developed an automated labeling solution for driving data, completing business data adaptation and designing prompts for model inference. The project was packaged as a scalable operator to boost large-scale data production.
- Optimized a 3D occupancy-control generation model for multi-modal driving data. This involved business data adaptation and model tuning for accurate 3D scene reconstruction.

Honors and Awards

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| ➤ Chongqing University First-Class Scholarship | Nov. 2024 |
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| ➤ Chongqing University Merit Student | Oct. 2024 |
| ➤ National Innovation and Entrepreneurship Training Program, Second-Level Completion | Jun. 2022 |

Professional Skills

- **Language skills:** English: CET-6 (515); Chinese (native)
- **Computer skills:** Microsoft Office, SQL, Python, Matlab
- **Professional skills:** Python (PyTorch, OpenCV), C/C++, Java, LaTeX